

CHARAN ELAMPARITHI KALA

Co-op Education Student

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SUMMARY OF SKILLS

- Proficient in **SQL (PostgreSQL)** through production database design, including schema architecture, automated calculation logic, and tiebreaker resolution across deployed applications.
- Hands-on **AI/ML and Generative AI** exposure: implemented transfer learning (VGG16, ResNet50) achieving 89.2% accuracy; roadmapping GenAI scheduling features in a live production app.
- Practical **Agile** experience through sprint planning, stand-ups, UAT, and requirements gathering as a Research Assistant; strong technical documentation and stakeholder communication skills.
- Full-stack developer with production deployments in **Python, TypeScript, SQL**, and MS Office 365, with experience translating complex business requirements into reliable, data-driven outputs.

TECHNICAL SKILLS

Programming & Data: Python, SQL (PostgreSQL), JavaScript, TypeScript, HTML/CSS

Analytics & AI/ML: Transfer learning (VGG16, ResNet50), SVM, Logistic Regression, AI-assisted development (Claude Code), introductory GenAI

Tools & Platforms: MS Office 365 (Excel, PowerPoint, Word, Outlook), Git/GitHub, Next.js, Supabase

Methodologies: Agile, Scrum, SDLC, UAT, requirements gathering, project documentation

EDUCATION

Bachelor of Computer Information Systems

Expected April 2028

University of the Fraser Valley — Abbotsford, BC

- GPA: 3.74 | Dean's List – Fall 2025 & Winter 2026 | Coursework: Database & DBMS, Agile SDLC, Web Dev, Networking & Security, Project Management, Machine Learning (COMP 381)

WORK EXPERIENCE

Founder & Instructor — S.T.A.C.K. (Online CS Tutoring Platform)

Aug 2025 – Present

Remote

- Designed and delivered a 10+ hour Python Bootcamp, developing curriculum, assessments, and progress reports while communicating outcomes to parents and student stakeholders.

Research Assistant

Oct 2024 – Present

University of the Fraser Valley — Abbotsford, BC

- Contribute to Agile sprint planning and stand-ups; conduct UAT on programming exercises, documenting defects and producing technical procedures that improve knowledge transfer and team efficiency.
- Collaborate cross-functionally with faculty to gather requirements and translate educational needs into technical solutions on PrairieLearn.

Assistant Coach

Mar 2024 – Present

Pursue Badminton — Surrey, BC

- Coordinate scheduling and lead sessions for 50+ students; manage administrative tasks and report outcomes to program leadership, demonstrating strong organizational and communication skills.

PROJECTS

Pre-Op Medication Reference • Next.js, TypeScript, Supabase, PostgreSQL

May 2026 – Present

Solo Project (In Development)

- Building a full-stack clinical decision support tool featuring role-based auth (Doctor/Admin), a Supabase-backed drug database with admin CRUD, and procedure date calculations with urgency color-coding across 35+ medications.

Badminton Tournament Manager • Next.js, TypeScript, PostgreSQL, Prisma

Mar 2026 – May 2026

Solo Project (Deployed) | github.com/CharanElam | easy-badminton.onrender.com

- Designed and deployed a data-driven tournament management app for a real BC event, architecting a PostgreSQL schema for standings, seeding, and tiebreaker resolution across multiple tournament formats.
- Leveraged AI-assisted development (Claude Code) throughout; roadmapping GenAI features including an intelligent scheduling assistant and match optimizer.

Breast Cancer Classification • Python, VGG16, ResNet50, SVM

May 2025 – Aug 2025

COMP 381 — Machine Learning | University of the Fraser Valley

- Classified breast ultrasound images as benign or malignant using transfer learning combined with traditional ML classifiers on a 647-image dataset, achieving 89.2% accuracy.
- Produced a full project report and presentation communicating model selection, performance metrics, and clinical implications to a non-technical academic audience.